<u>**Report Assignment 3**</u>

<u>GITHUB</u>

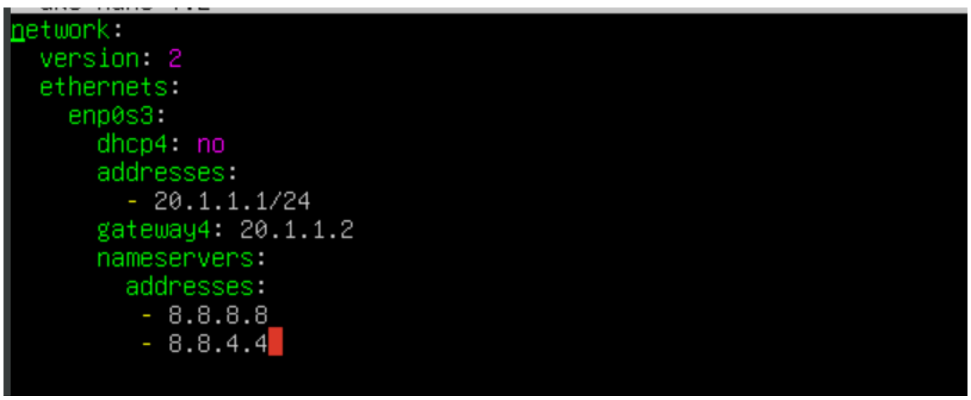
Vimansh Mahajan
2022572

Question 1 Summary

Objective: Set up four Virtual Machines (VMs) in a network configuration as described in the assignment figure.

Command used: sudo nano /etc/netplan/01-netcfg.yaml

CLIENT



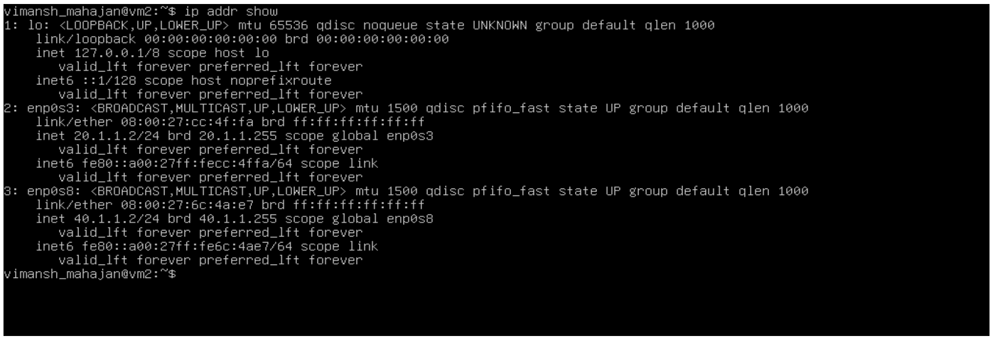
```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 20.1.1.1/24
      gateway4: 20.1.1.2
      nameservers:
        addresses:
         - 8.8.8.8
         - 8.8.4.4
```

```
vimansh_mahajan@vm1:~$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:b4:1f:1f brd ff:ff:ff:ff:ff:ff
    inet 20.1.1.1/24 brd 20.1.1.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 86235sec preferred_lft 86235sec
    inet6 fe80::a00:27ff:feb4:1f1f/64 scope link
       valid_lft forever preferred_lft forever
vimansh_mahajan@vm1:~$ _
```

inet : 20.1.1.1/24 for enp0s3

GATEWAY

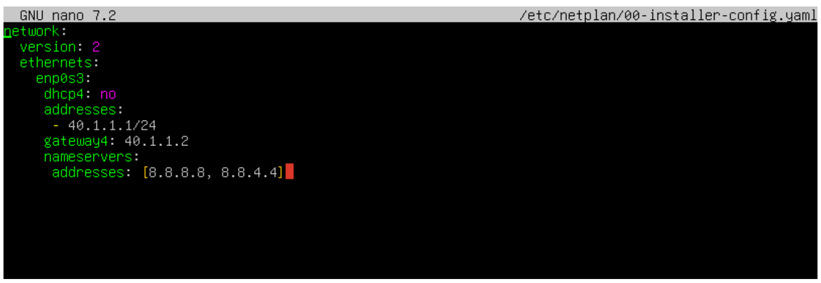
```
  GNU nano 7.2                                /etc/netplan/00-installer-config.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 20.1.1.2/24
    enp0s8:
     dhcp4: no
     addresses:
        - 40.1.1.2/24
```



```
vimansh_mahajan@vm2:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:cc:4f:fa brd ff:ff:ff:ff:ff:ff
    inet 20.1.1.2/24 brd 20.1.1.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fecc:4ffa/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:6c:4a:e7 brd ff:ff:ff:ff:ff:ff
    inet 40.1.1.2/24 brd 40.1.1.255 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe6c:4ae7/64 scope link
       valid_lft forever preferred_lft forever
vimansh_mahajan@vm2:~$
```

Inet: 20.1.1.2/24 for enp0s3(client and gateway) and 40.1.1.2/24 for enp0s8 (gateway to server)

VM3 (SERVER 1)

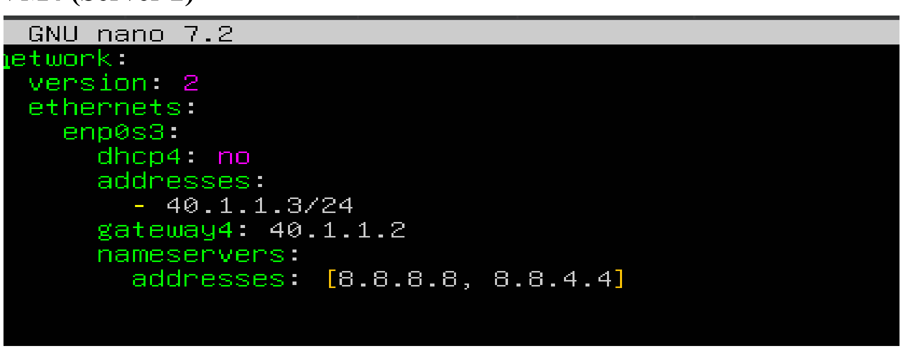


```
  GNU nano 7.2                                        /etc/netplan/00-installer-config.yaml
network:
  version: 2
  ethernets:
    enp0s3:
     dhcp4: no
     addresses:
      - 40.1.1.1/24
     gateway4: 40.1.1.2
     nameservers:
      addresses: [8.8.8.8, 8.8.4.4]
```

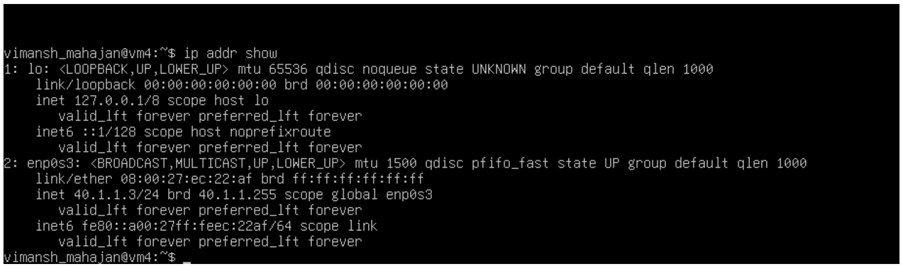
```
vimansh_mahajan@vm3:~$ ip addr show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:9c:49:a9 brd ff:ff:ff:ff:ff:ff
    inet 40.1.1.1/24 brd 40.1.1.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 86330sec preferred_lft 86330sec
    inet6 fe80::a00:27ff:fe9c:49a9/64 scope link
       valid_lft forever preferred_lft forever
vimansh_mahajan@vm3:~$ _
```

inet : 40.1.1.1/24 for enp0s3

VM4 (Server 2)



```
  GNU nano 7.2
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 40.1.1.3/24
      gateway4: 40.1.1.2
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```



inet : 40.1.1.3/24 for enp0s3

Commands:
sudo ip route add 40.1.1.0/24 via 20.1.1.2 (at client):

Adds a route on the client machine, specifying that packets with a destination IP in the 40.1.1.0/24 subnet should be routed via the gateway 20.1.1.2

sudo ip route add 20.1.1.0/24 via 40.1.1.2 (server 1):

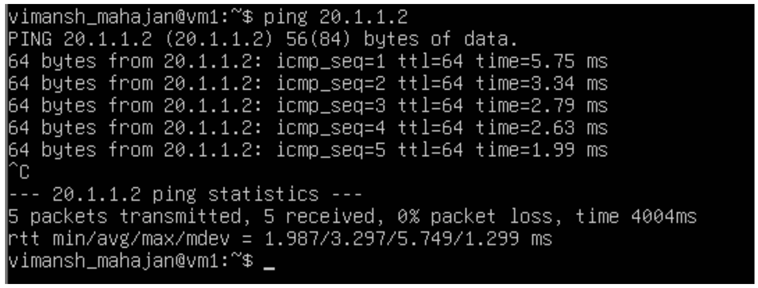
Adds a route on Server 1, directing packets bound for the 20.1.1.0/24 subnet to go through the gateway 40.1.1.2.

sudo ip route add 20.1.1.0/24 via 40.1.1.2 (server 2):

Adds a route on Server 2, similarly specifying that packets destined for the 20.1.1.0/24 subnet should be routed via 40.1.1.2.

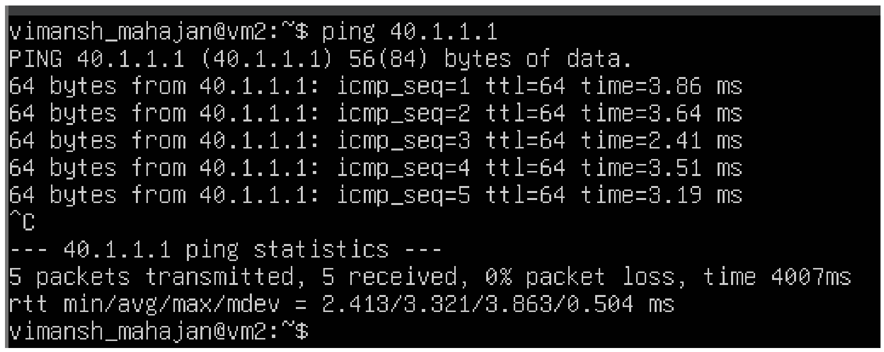
Steps Taken:

1. **VM Configuration**:
    - **VM1**:
        - IP Address: 20.1.1.1/24
        - Network Interface: enp0s3
    - **VM2**:
        - IP Address: 20.1.1.2/24 (Client-Gateway Interface: enp0s3)
        - Gateway Interface: enp0s8 with IP Address 40.1.1.2/24 (Gateway-Server Interface)
    - **VM3** (Server 1):
        - IP Address: 40.1.1.1/24
        - Network Interface: enp0s3
    - **VM4** (Server 2):
        - IP Address: 40.1.1.3/24
        - Network Interface: enp0s3
2. **Network Testing**:
    - Verified connectivity between VMs using ping commands:
        - **VM1 to VM2**: Successful



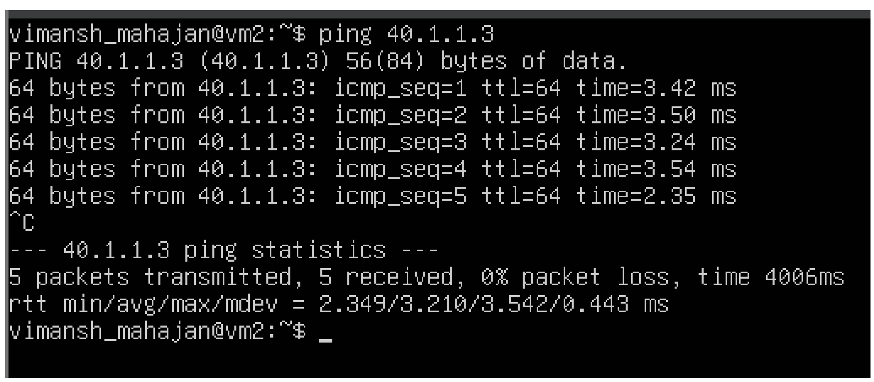
```
vimansh_mahajan@vm1:~$ ping 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=64 time=5.75 ms
64 bytes from 20.1.1.2: icmp_seq=2 ttl=64 time=3.34 ms
64 bytes from 20.1.1.2: icmp_seq=3 ttl=64 time=2.79 ms
64 bytes from 20.1.1.2: icmp_seq=4 ttl=64 time=2.63 ms
64 bytes from 20.1.1.2: icmp_seq=5 ttl=64 time=1.99 ms
^C
--- 20.1.1.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 1.987/3.297/5.749/1.299 ms
vimansh_mahajan@vm1:~$ _
```

- **VM2 to VM3**: Successful



```
vimansh_mahajan@vm2:~$ ping 40.1.1.1
PING 40.1.1.1 (40.1.1.1) 56(84) bytes of data.
64 bytes from 40.1.1.1: icmp_seq=1 ttl=64 time=3.86 ms
64 bytes from 40.1.1.1: icmp_seq=2 ttl=64 time=3.64 ms
64 bytes from 40.1.1.1: icmp_seq=3 ttl=64 time=2.41 ms
64 bytes from 40.1.1.1: icmp_seq=4 ttl=64 time=3.51 ms
64 bytes from 40.1.1.1: icmp_seq=5 ttl=64 time=3.19 ms
^C
--- 40.1.1.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 2.413/3.321/3.863/0.504 ms
vimansh_mahajan@vm2:~$
```

- **VM2 to VM4**: Successful



```
vimansh_mahajan@vm2:~$ ping 40.1.1.3
PING 40.1.1.3 (40.1.1.3) 56(84) bytes of data.
64 bytes from 40.1.1.3: icmp_seq=1 ttl=64 time=3.42 ms
64 bytes from 40.1.1.3: icmp_seq=2 ttl=64 time=3.50 ms
64 bytes from 40.1.1.3: icmp_seq=3 ttl=64 time=3.24 ms
64 bytes from 40.1.1.3: icmp_seq=4 ttl=64 time=3.54 ms
64 bytes from 40.1.1.3: icmp_seq=5 ttl=64 time=2.35 ms
^C
--- 40.1.1.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 2.349/3.210/3.542/0.443 ms
vimansh_mahajan@vm2:~$ _
```

(b) Gateway configuration

Enable IP Forwarding: sudo sysctl -w net.ipv4.ip_forward=1

This command enables IP forwarding in the kernel. It allows VM2 to act as a router, forwarding packets between interfaces (enp0s3 and enp0s8).

Add Forwarding Rules: sudo iptables -A FORWARD -i enp0s8 -o enp0s3 -j ACCEPT

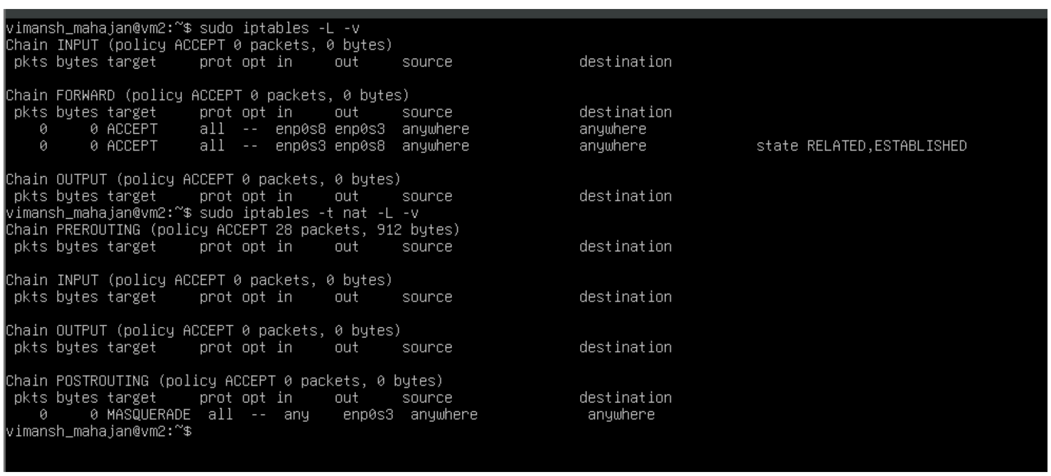
This rule accepts (-j ACCEPT) any traffic coming in from the server network (enp0s8) and forwards it to the client network (enp0s3).

Allow traffic from clients to servers: sudo iptables -A FORWARD -i enp0s3 -o enp0s8 -m state --state ESTABLISHED,RELATED -j ACCEPT

This rule allows traffic to return from clients to the server network, provided it's part of an already established connection (-m state --state ESTABLISHED,RELATED).

Network Address Translation (NAT): sudo iptables -t nat -A POSTROUTING -o enp0s3 -j MASQUERADE

This rule uses **NAT** to masquerade outgoing traffic from the server network (enp0s8). It modifies the packets' source IP address to appear as coming from VM2, allowing correct routing back to the clients.



```
vimansh_mahajan@vm2:~$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     all  --  enp0s8 enp0s3  anywhere             anywhere
    0     0 ACCEPT     all  --  enp0s3 enp0s8  anywhere             anywhere             state RELATED,ESTABLISHED

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
vimansh_mahajan@vm2:~$ sudo iptables -t nat -L -v
Chain PREROUTING (policy ACCEPT 28 packets, 912 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 MASQUERADE  all  --  any    enp0s3  anywhere             anywhere
vimansh_mahajan@vm2:~$
```

GATEWAY

Question 2 summary:

Traffic Filtering at the Gateway VM

(a) Blocking All Traffic (Except for Ping) Destined to the Server 40.1.1.1/24

1. The goal was to configure the gateway (VM2) to block all traffic destined for the server at IP address 40.1.1.1/24, except for ICMP (ping) requests. This ensures that the server can only respond to ping requests while blocking all other types of traffic.
2. **Implementation Steps**:
    - Applied the following iptables rules to achieve the desired traffic filtering
    1) **sudo iptables -A FORWARD -d 40.1.1.1/24 -p icmp -j ACCEPT:**

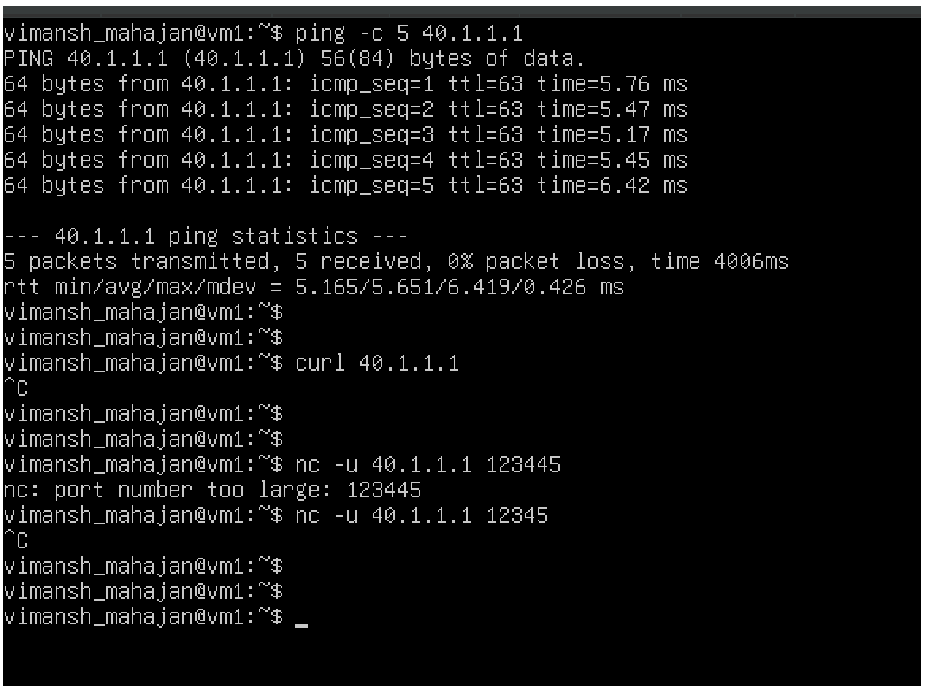
Allow ICMP (ping) traffic

2) **sudo iptables -A FORWARD -d 40.1.1.1/24 -j DROP**

Block all other traffic to the server

Verification:

- Conducted a ping test from the client (VM1) to the server (40.1.1.1), which returned successful replies, confirming that ICMP traffic was allowed.
- Executed a curl command to connect to the server's HTTP port: curl 40.1.1.1, The command returned an error indicating that it could not connect to the server, confirming that all non-ping traffic was effectively blocked. (TCP traffic got blocked)
- Executed nc -u 40.1.1.1 which got blocked, so UDP traffic got blocked



```
vimansh_mahajan@vm1:~$ ping -c 5 40.1.1.1
PING 40.1.1.1 (40.1.1.1) 56(84) bytes of data.
64 bytes from 40.1.1.1: icmp_seq=1 ttl=63 time=5.76 ms
64 bytes from 40.1.1.1: icmp_seq=2 ttl=63 time=5.47 ms
64 bytes from 40.1.1.1: icmp_seq=3 ttl=63 time=5.17 ms
64 bytes from 40.1.1.1: icmp_seq=4 ttl=63 time=5.45 ms
64 bytes from 40.1.1.1: icmp_seq=5 ttl=63 time=6.42 ms

--- 40.1.1.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 5.165/5.651/6.419/0.426 ms
vimansh_mahajan@vm1:~$
vimansh_mahajan@vm1:~$
vimansh_mahajan@vm1:~$ curl 40.1.1.1
^C
vimansh_mahajan@vm1:~$
vimansh_mahajan@vm1:~$
vimansh_mahajan@vm1:~$ nc -u 40.1.1.1 123445
nc: port number too large: 123445
vimansh_mahajan@vm1:~$ nc -u 40.1.1.1 12345
^C
vimansh_mahajan@vm1:~$
vimansh_mahajan@vm1:~$
vimansh_mahajan@vm1:~$ _
```

- Ping -c 5 40.1.1.1 worked successfully.
- Curl 40.1.1.1 (TCP protocol) failed to produce any output (got blocked)
- nc -u (for udp) 40.1.1.1 12345 (UDP) failed to produce any output (got blocked)

(b)

The aim was to configure the gateway to block only TCP traffic initiated by the client (20.1.1.1/24), while allowing other types of traffic from this IP.

Implementation Steps:

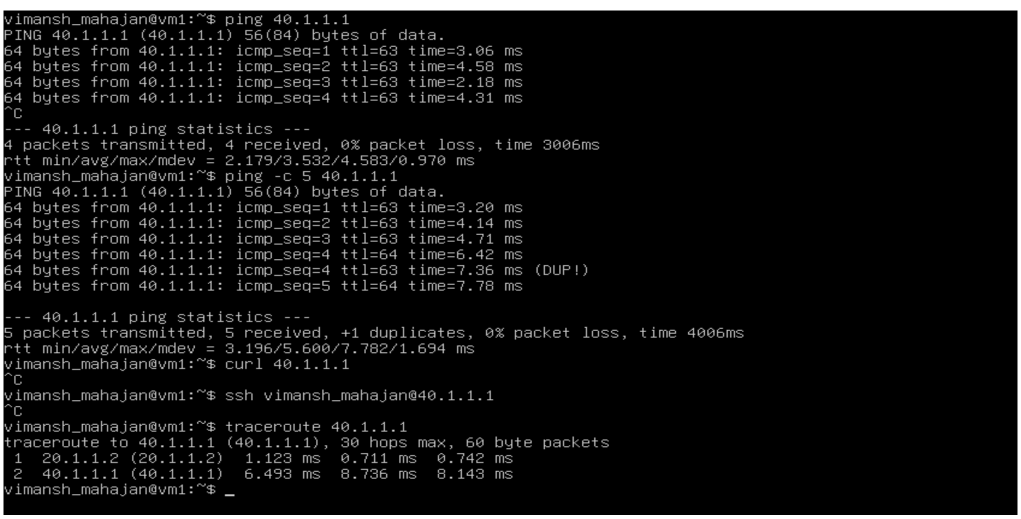
On the gateway VM (VM2), applied the following iptables rule:

sudo iptables -A FORWARD -s 20.1.1.1/24 -p tcp -j DROP # Block TCP traffic from VM1

- This rule explicitly targets TCP traffic initiated from the source IP of 20.1.1.1/24 and drops it.
- -A FORWARD: Appends this rule to the FORWARD chain, affecting packets that are routed through the gateway.
- -s 20.1.1.1/24: Specifies the source IP to block
- -p gives protocol that is in our case tcp
- -j DROP: Drops packets that match this rule without sending any response.

Verification:

- Ping 40.1.1.1 and traceroute 40.1.1.1 worked successfully, indicating that non-TCP traffic is not blocked
- Curl 40.1.1.1 and ssh vimansh_mahajan@40.1.1.1 which are based on TCP did not work as traffic was blocked


```
vimansh_mahajan@vm1:~$ ping 40.1.1.1
PING 40.1.1.1 (40.1.1.1) 56(84) bytes of data.
64 bytes from 40.1.1.1: icmp_seq=1 ttl=63 time=3.06 ms
64 bytes from 40.1.1.1: icmp_seq=2 ttl=63 time=4.58 ms
64 bytes from 40.1.1.1: icmp_seq=3 ttl=63 time=2.18 ms
64 bytes from 40.1.1.1: icmp_seq=4 ttl=63 time=4.31 ms
^C
--- 40.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 2.179/3.532/4.583/0.970 ms
vimansh_mahajan@vm1:~$ ping -c 5 40.1.1.1
PING 40.1.1.1 (40.1.1.1) 56(84) bytes of data.
64 bytes from 40.1.1.1: icmp_seq=1 ttl=63 time=3.20 ms
64 bytes from 40.1.1.1: icmp_seq=2 ttl=63 time=4.14 ms
64 bytes from 40.1.1.1: icmp_seq=3 ttl=63 time=4.71 ms
64 bytes from 40.1.1.1: icmp_seq=4 ttl=64 time=6.42 ms
64 bytes from 40.1.1.1: icmp_seq=4 ttl=63 time=7.36 ms (DUP!)
64 bytes from 40.1.1.1: icmp_seq=5 ttl=64 time=7.78 ms

--- 40.1.1.1 ping statistics ---
5 packets transmitted, 5 received, +1 duplicates, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 3.196/5.600/7.782/1.694 ms
vimansh_mahajan@vm1:~$ curl 40.1.1.1
^C
vimansh_mahajan@vm1:~$ ssh vimansh_mahajan@40.1.1.1
^C
vimansh_mahajan@vm1:~$ traceroute 40.1.1.1
traceroute to 40.1.1.1 (40.1.1.1), 30 hops max, 60 byte packets
 1  20.1.1.2 (20.1.1.2)  1.123 ms  0.711 ms  0.742 ms
 2  40.1.1.1 (40.1.1.1)  6.493 ms  8.736 ms  8.143 ms
vimansh_mahajan@vm1:~$ _
```

Similarly, for 40.1.1.3:

```
vimansh_mahajan@vm1:~$ ping 40.1.1.3
PING 40.1.1.3 (40.1.1.3) 56(84) bytes of data.
64 bytes from 40.1.1.3: icmp_seq=1 ttl=63 time=5.10 ms
64 bytes from 40.1.1.3: icmp_seq=2 ttl=63 time=5.24 ms
64 bytes from 40.1.1.3: icmp_seq=3 ttl=63 time=6.16 ms
64 bytes from 40.1.1.3: icmp_seq=4 ttl=63 time=5.06 ms
64 bytes from 40.1.1.3: icmp_seq=5 ttl=63 time=5.27 ms
^C
--- 40.1.1.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 5.063/5.365/6.159/0.404 ms
vimansh_mahajan@vm1:~$ curl 40.1.1.3
^C
vimansh_mahajan@vm1:~$ ssh vimansh_mahajan@40.1.1.3
^C
vimansh_mahajan@vm1:~$ _
```

Question 3:

(a)

UDP (First picture is 20.1.1.1, second is 40.1.1.3)

```
vimansh_mahajan@vm4:~$ iperf -s -u
------------------------------------------------------------
Server listening on UDP port 5001
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 40.1.1.3 port 5001 connected with 20.1.1.1 port 59330
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-10.0039 sec  12.5 MBytes  10.5 Mbits/sec   0.258 ms 0/8920 (0%)
^Cvimansh_mahajan@vm4:~$ _
```

```
vimansh_mahajan@vm1:~$ iperf -c 40.1.1.3 -u -b 10M
------------------------------------------------------------
Client connecting to 40.1.1.3, UDP port 5001
Sending 1470 byte datagrams, IPG target: 1121.52 us (kalman adjust)
UDP buffer size:  208 KByte (default)
------------------------------------------------------------
[  1] local 20.1.1.1 port 59330 connected with 40.1.1.3 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0026 sec  12.5 MBytes  10.5 Mbits/sec
[  1] Sent 8921 datagrams
[  1] Server Report:
[ ID] Interval       Transfer     Bandwidth        Jitter   Lost/Total Datagrams
[  1] 0.0000-10.0039 sec  12.5 MBytes  10.5 Mbits/sec   0.258 ms 0/8920 (0%)
vimansh_mahajan@vm1:~$ ^C
vimansh_mahajan@vm1:~$
```

Client (First Image):

- **Command Used**: iperf -c 40.1.1.3 -u -b 10M
    - -c 40.1.1.3: Connect to the server at IP 40.1.1.3.
    - -u: Use UDP instead of TCP.
    - -b 10M: Set the target bandwidth to 10 Mbps.
- **Test Details**:
    - The client sent 1470-byte datagrams, with an inter-packet gap of 1121.52 microseconds
    - **Transfer**: 12.5 MB (Megabytes) of data sent during the test.
    - **Bandwidth Achieved**: 10.5 Mbps, slightly higher than the target bandwidth of 10 Mbps.
    - **Jitter**: 0.258 ms (milliseconds). Jitter refers to the variation in time delay between data packets, and in this case, it's very low, indicating a stable connection.
    - **Lost/Total Datagrams**: 0/8920 (0% loss). This indicates no packet loss, which is ideal for UDP-based communication.

Server (Second Image):

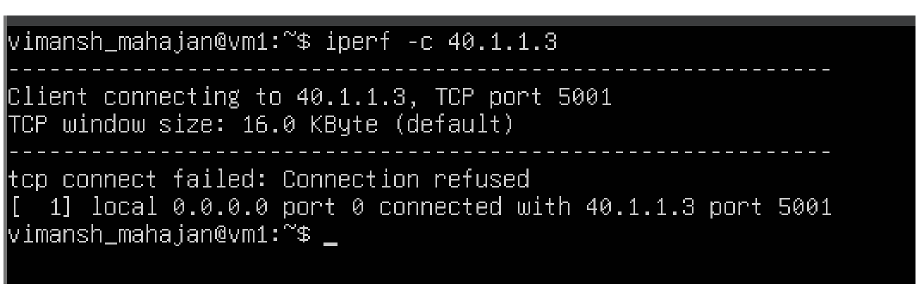
- **Command Used**: iperf -s -u
    - -s: Run the server in listening mode.
    - -u: Listen for UDP traffic.
- **Test Details**:
    - The server received the 12.5 MB of data sent by the client.
    - **Bandwidth Achieved**: 10.5 Mbps, matching the client-side results.
    - **Jitter**: 0.258 ms, identical to the client report.
    - **Lost/Total Datagrams**: 0/8920, meaning no packet loss.

Conclusion:

The iperf test shows a stable and reliable UDP connection between the client and server:

- The actual bandwidth achieved is slightly above the 10 Mbps target (10.5 Mbps).
- The jitter is low (0.258 ms), indicating minimal variation in packet arrival times.
- There is no packet loss (0%), meaning that all packets were successfully delivered without loss or errors.

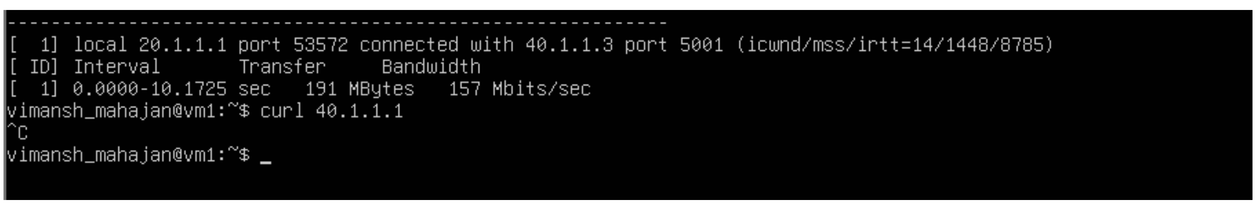
TCP (First picture is 20.1.1.1, second is 40.1.1.3)



To verify that the gateway correctly blocks TCP traffic originating from 20.1.1.1/24, I tested TCP bandwidth using the iperf tool between 20.1.1.1 (client) and 40.1.1.3 (server). The firewall rules implemented on the gateway are configured to reject only TCP traffic initiated by 20.1.1.1/24. Consequently, when attempting to establish a TCP connection, the firewall intercepted and dropped the packets, leading to a “Connection Refused” error.

This outcome confirms that the rule is effectively blocking TCP packets as specified, while allowing other types of traffic. Thus, the iptables configuration has successfully restricted only the desired traffic type, demonstrating selective blocking functionality at the gateway.

Note: with command- sudo iptables -A INPUT -s 20.1.1.1/24 -p tcp -j DROP i.e. adding drop rule to input chain although the traffic of tcp gets blocked the TCP bandwidth can still be measured.



At client:

For interval of 0 to 10.1725 secs, 191 MBytes of data is transferred with a bandwidth of 157 Mbits/sec.

At server (40.1.1.3):

for interval of 0 to 10.1484 secs, 191MBytes of data transfer takes place with a bandwidth of 158 Mbits/sec

(b)

```
vimansh_mahajan@vm1:~$ ping -c 10 40.1.1.1
PING 40.1.1.1 (40.1.1.1) 56(84) bytes of data.
64 bytes from 40.1.1.1: icmp_seq=1 ttl=63 time=4.68 ms
64 bytes from 40.1.1.1: icmp_seq=2 ttl=63 time=4.11 ms
64 bytes from 40.1.1.1: icmp_seq=3 ttl=63 time=6.47 ms
64 bytes from 40.1.1.1: icmp_seq=4 ttl=63 time=4.82 ms
64 bytes from 40.1.1.1: icmp_seq=5 ttl=63 time=6.14 ms
64 bytes from 40.1.1.1: icmp_seq=6 ttl=63 time=5.40 ms
64 bytes from 40.1.1.1: icmp_seq=7 ttl=63 time=5.22 ms
64 bytes from 40.1.1.1: icmp_seq=8 ttl=63 time=4.70 ms
64 bytes from 40.1.1.1: icmp_seq=9 ttl=63 time=4.30 ms
64 bytes from 40.1.1.1: icmp_seq=10 ttl=63 time=5.98 ms

--- 40.1.1.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 4.106/5.180/6.465/0.760 ms
vimansh_mahajan@vm1:~$
```

```
vimansh_mahajan@vm1:~$ ping -c 10 40.1.1.3
PING 40.1.1.3 (40.1.1.3) 56(84) bytes of data.
64 bytes from 40.1.1.3: icmp_seq=1 ttl=63 time=9.21 ms
64 bytes from 40.1.1.3: icmp_seq=2 ttl=63 time=5.20 ms
64 bytes from 40.1.1.3: icmp_seq=3 ttl=63 time=4.83 ms
64 bytes from 40.1.1.3: icmp_seq=4 ttl=63 time=5.22 ms
64 bytes from 40.1.1.3: icmp_seq=5 ttl=63 time=5.96 ms
64 bytes from 40.1.1.3: icmp_seq=6 ttl=63 time=4.22 ms
64 bytes from 40.1.1.3: icmp_seq=7 ttl=63 time=6.16 ms
64 bytes from 40.1.1.3: icmp_seq=8 ttl=63 time=5.11 ms
64 bytes from 40.1.1.3: icmp_seq=9 ttl=63 time=5.20 ms
64 bytes from 40.1.1.3: icmp_seq=10 ttl=63 time=5.52 ms

--- 40.1.1.3 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9015ms
rtt min/avg/max/mdev = 4.220/5.663/9.213/1.290 ms
vimansh_mahajan@vm1:~$
```

RTT Results:

- **From 20.1.1.1 to 40.1.1.1**:
  - **Minimum RTT**: 4.106 ms
  - **Average RTT**: 5.180 ms
  - **Maximum RTT**: 6.465 ms
  - **mDEV**: 0.760 ms
- **From 20.1.1.1 to 40.1.1.3**:
  - **Minimum RTT**: 4.220 ms
  - **Average RTT**: 5.663 ms
  - **Maximum RTT**: 9.213 ms
  - **mDEV**: 1.290 ms

Analysis:

Minimum RTT:

- The minimum RTT is slightly better for the path from **20.1.1.1 to 40.1.1.1** (4.106 ms vs. 4.220 ms). This suggests that the first route has a marginally faster direct path, possibly due to lower initial network overhead.

Average RTT:

- The average RTT for the **20.1.1.1 to 40.1.1.1** path is **5.180 ms**, which is lower than the **5.663 ms** observed for the **20.1.1.1 to 40.1.1.3** path. This indicates that, on average, the connection to **40.1.1.1** is quicker and more efficient, suggesting lower latency between these two endpoints compared to the route to **40.1.1.3**.

Maximum RTT:

- The **maximum RTT** from **20.1.1.1 to 40.1.1.1** is **6.465 ms**, significantly lower than the **9.213 ms** to **40.1.1.3**. This indicates that the second route (to **40.1.1.3**) has larger spikes in latency, which could be caused by temporary congestion, network path inefficiency.

mDEV (Mean Deviation):

- The **mDEV** value, which shows the variation or consistency in RTT, is smaller for the route from **20.1.1.1 to 40.1.1.1** (**0.760 ms**) compared to the route to **40.1.1.3** (**1.290 ms**). This suggests that the first route is more stable with less variability in round-trip times, while the second route experiences more fluctuations in network performance.

No significant differences reason:

- **Physical Distance and Infrastructure**: Since both servers are at the same physical location i.e. have similar connection paths, the length of the network path is virtually identical. Thus, the minor differences in RTT can be attributed to transient network conditions rather than any inherent distance or path inefficiency.
- **Network Configuration and Equipment**: The network devices, such as routers and switches, along the path from the client to each server are configured the same way. Both routes should experience similar processing and queuing times at these devices.
- **Traffic Consistency**: Given equal network traffic conditions, both paths are subjected to similar levels of load and congestion. The observed variations in RTT, therefore, likely stem from minor, short-lived network fluctuations rather than significant architectural differences.

Question 4:

(a) And (b)

Change the source IP address of every packet from 20.1.1.1/24 to 40.1.1.2/24:

sudo iptables -t nat -A POSTROUTING -s 20.1.1.1/24 -o enp0s8 -j SNAT --to-source 40.1.1.2

- -t nat: Use the nat (Network Address Translation) table.
- -A POSTROUTING: Append a rule to the POSTROUTING chain, which manipulates packets just before they leave the gateway.
- -s 20.1.1.1/24: Specifies the source IP range to match packets coming from 20.1.1.1/24.
- -o enp0s8: Specifies the outgoing interface (from gateway to servers).
- -j SNAT: Jumps to the SNAT target, which allows source address translation.
- --to-source 40.1.1.2: Translates the source IP to 40.1.1.2 for packets going out.

Revert the Destination IP Address for Responses:

When the response for the packet arrives at the gateway, revert the destination IP back to the original source (20.1.1.1/24):

sudo iptables -t nat -A PREROUTING -d 40.1.1.2 -i enp0s8 -j DNAT --to-destination 20.1.1.1

- -t nat: Use the nat table.
- -A PREROUTING: Append a rule to the PREROUTING chain, which processes packets as they arrive at the gateway before making routing decisions.
- -d 40.1.1.2: Specifies the destination IP to match the response packets directed to 40.1.1.2.
- -i enp0s8: Specifies the incoming interface (from server to gateway).
- -j DNAT: Jumps to the DNAT target, which changes the destination address.
- --to-destination 20.1.1.1: Translates the destination address back to 20.1.1.1, ensuring the response reaches the original client.

```
vimansh_mahajan@vm2:~$ sudo iptables -t nat -L -v
Chain PREROUTING (policy ACCEPT 421 packets, 18781 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DNAT       icmp --  any    any     20.1.1.0/24          anywhere             to:40.1.1.2

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 4 packets, 324 bytes)
 pkts bytes target     prot opt in     out     source               destination
  106  7418 MASQUERADE  all  --  any    enp0s3  anywhere             anywhere
    0     0 SNAT       icmp --  any    any     anywhere             40.1.1.2             to:20.1.1.1
vimansh_mahajan@vm2:~$
```

Under Chain PREROUTING: from source 20.1.1.0/24, when packet is forwarded its IP address is changed to 40.1.1.2, as visible in the given section

Under Chain POSTROUTING: when packet response for the packet arrives at the gateway, the destination IP address is changed to the original.

Tcpdump at client

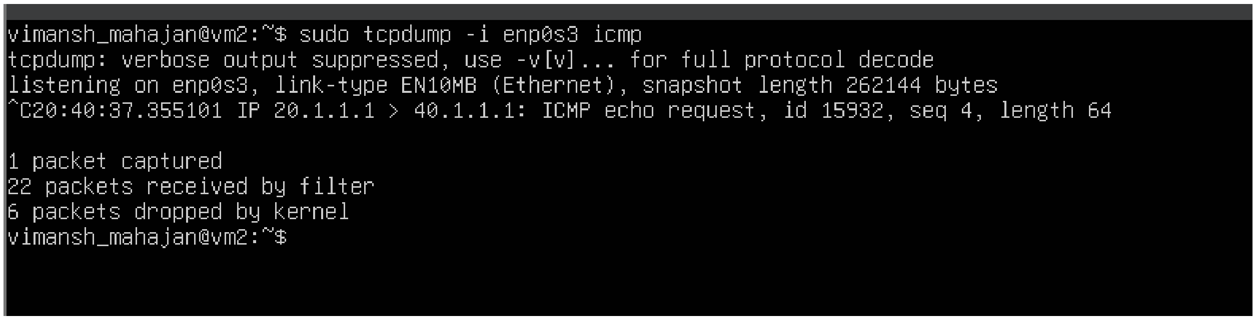
```
vimansh_mahajan@vm1:~$ sudo tcpdump -i enp0s3 -n -v icmp
[sudo] password for vimansh_mahajan:
tcpdump: listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
20:57:29.776747 IP (tos 0x0, ttl 64, id 31520, offset 0, flags [DF], proto ICMP (1), length 84)
    20.1.1.1 > 40.1.1.1: ICMP echo request, id 15961, seq 1, length 64
20:57:29.779028 IP (tos 0x0, ttl 64, id 52137, offset 0, flags [none], proto ICMP (1), length 84)
    40.1.1.1 > 20.1.1.1: ICMP echo reply, id 15961, seq 1, length 64
^C
2 packets captured
2 packets received by filter
0 packets dropped by kernel
vimansh_mahajan@vm1:~$
```

<u>Ping from the client to 40.1.1.1</u>

```
vimansh_mahajan@vm1:~$ ping 40.1.1.1 -c 1
PING 40.1.1.1 (40.1.1.1) 56(84) bytes of data.
64 bytes from 40.1.1.1: icmp_seq=1 ttl=64 time=1.82 ms

--- 40.1.1.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.823/1.823/1.823/0.000 ms
vimansh_mahajan@vm1:~$ 20:56:36.580765 IP vm1 > 40.1.1.1: ICMP echo request, id 15956, seq 1, length 64
20:56:36.582541 IP 40.1.1.1 > vm1: ICMP echo reply, id 15956, seq 1, length 64
ping 40.1.1.1 -c 1
PING 40.1.1.1 (40.1.1.1) 56(84) bytes of data.
64 bytes from 40.1.1.1: icmp_seq=1 ttl=64 time=2.33 ms

--- 40.1.1.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2.331/2.331/2.331/0.000 ms
vimansh_mahajan@vm1:~$ 20:57:29.776747 IP vm1 > 40.1.1.1: ICMP echo request, id 15961, seq 1, length 64
20:57:29.779028 IP 40.1.1.1 > vm1: ICMP echo reply, id 15961, seq 1, length 64
```

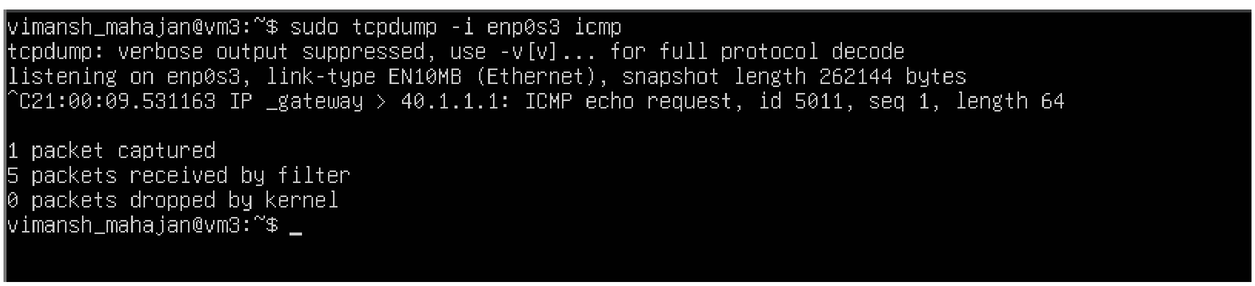
When ping 40.1.1.1 is run from the client, then tcpdump at client captures the above-given statistics, which shows that the client sees the response arriving from the server at 40.1.1.1

Tcpdump at gateway (20.1.1.2 or interface enp0s3)



Tcpdump at gateway (40.1.1.2 or interface enp0s8)

Tcpdump at 40.1.1.1 (server)



As we can see front the above two screenshots that gateway receives packet from the client (20.1.1.1) at its interface (40.1.1.1) but before forwarding it to 40.1.1.2 the outgoing packet IP is changed to 40.1.1.2. The packet received at server (40.1.1.1 or vm3) is also from 40.1.1.2 or gateway.

Part (b): Revert the destination IP address to the original

```
vimansh_mahajan@vm2:~$ sudo tcpdump -i enp0s8 icmp[icmptype] = icmp-echoreply
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s8, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C21:07:20.894452 IP 40.1.1.1 > 40.1.1.2: ICMP echo reply, id 5034, seq 1, length 64

1 packet captured
1 packet received by filter
0 packets dropped by kernel
vimansh_mahajan@vm2:~$
```

The response received at enp0s8 from server if sent from server (40.1.1.1) to 40.1.1.2 (gateway) depicting correctness of the applied changes

```
rtt min/avg/max/mdev = 2.198/2.198/2.198/0.000 ms
vimansh_mahajan@vm2:~$ sudo tcpdump -i enp0s3 icmp[icmptype] = icmp-echoreply
[sudo] password for vimansh_mahajan:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C21:12:44.771838 IP 40.1.1.1 > 20.1.1.1: ICMP echo reply, id 15981, seq 1, length 64

1 packet captured
1 packet received by filter
0 packets dropped by kernel
vimansh_mahajan@vm2:~$
```

At gateway enp0s3 (between client and gateway) we see the response changing back from 40.1.1.1 to 20.1.1.1

```
vimansh_mahajan@vm1:~$ sudo tcpdump -i enp0s3 -n -v icmp
tcpdump: listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
21:09:32.724245 IP (tos 0x0, ttl 64, id 32915, offset 0, flags [DF], proto ICMP (1), length 84)
    20.1.1.1 > 40.1.1.1: ICMP echo request, id 15977, seq 1, length 64
21:09:32.726029 IP (tos 0x0, ttl 64, id 31315, offset 0, flags [none], proto ICMP (1), length 84)
    40.1.1.1 > 20.1.1.1: ICMP echo reply, id 15977, seq 1, length 64
^C
```

```
21:09:32.726029 IP (tos 0x0, ttl 64, id 31315, offset 0, flags [none], proto ICMP (1), length 84)
    40.1.1.1 > 20.1.1.1: ICMP echo reply, id 15977, seq 1, length 64
^C
```

At client, it sees the reply coming back from server address 20.1.1.1, which is what was required in the section

Request Summary

1. **VM1 (Client: 20.1.1.1)**:
    - **Sends**: ICMP echo request
        - **Before**: Source IP: 20.1.1.1, Destination IP: 40.1.1.1
        - **After**: Source IP: 40.1.1.2, Destination IP: 40.1.1.1 (modified by gateway)

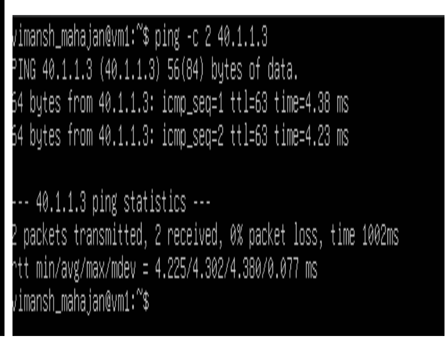
2. **VM2 (Gateway: 20.1.1.2)**:
    - **Receives**: ICMP echo request from VM1
        - **Changes Source**: From 20.1.1.1 to 40.1.1.2
        - **Forwards to VM3**: Source IP: 40.1.1.2, Destination IP: 40.1.1.1
3. **VM3 (Server: 40.1.1.1)**:
    - **Receives**: ICMP echo request from VM2

Response Summary

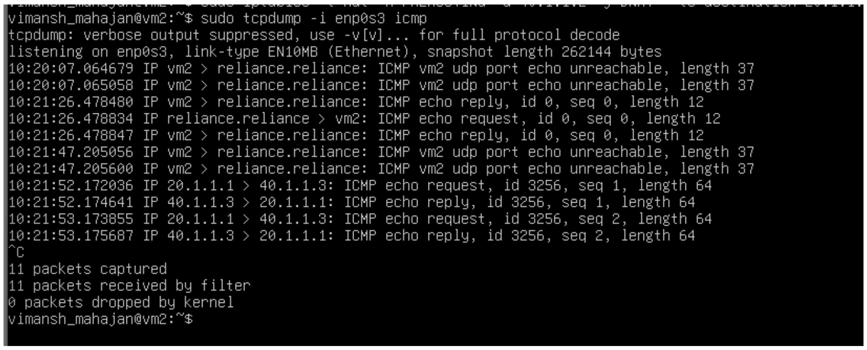
1. **VM3 (Server: 40.1.1.1)**:
    - **Sends**: ICMP echo reply
        - **Before Reply**: Source IP: 40.1.1.1, Destination IP: 40.1.1.2
2. **VM2 (Gateway: 20.1.1.2)**:
    - **Receives**: ICMP echo reply from VM3
        - **Reverts Destination**: From 40.1.1.2 back to 20.1.1.1
        - **Forwards to VM1**: Source IP: 40.1.1.1, Destination IP: 20.1.1.1
3. **VM1 (Client: 20.1.1.1)**:
    - **Receives**: ICMP echo reply from VM2
        - **Final**: Source IP: 40.1.1.1, Destination IP: 20.1.1.1

Similarly for ping between 20.1.1.1 and vm4 (40.1.1.3)

Client (20.1.1.1)

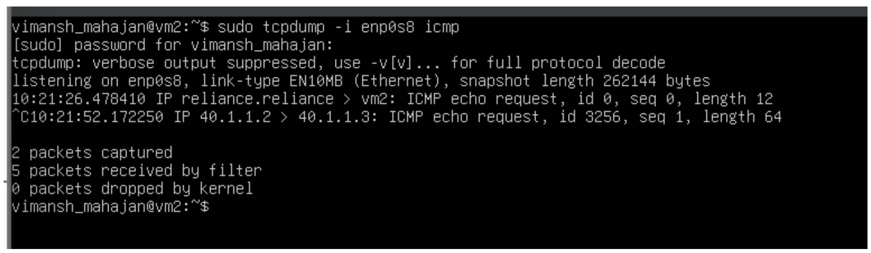


Gateway interface between client and gateway



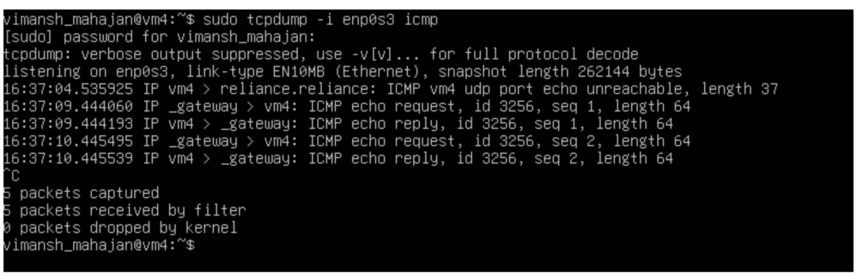
```
vimansh_mahajan@vm2:~$ sudo tcpdump -i enp0s3 icmp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:20:07.064679 IP vm2 > reliance.reliance: ICMP vm2 udp port echo unreachable, length 37
10:20:07.065058 IP vm2 > reliance.reliance: ICMP vm2 udp port echo unreachable, length 37
10:21:26.478480 IP vm2 > reliance.reliance: ICMP echo reply, id 0, seq 0, length 12
10:21:26.478834 IP reliance.reliance > vm2: ICMP echo request, id 0, seq 0, length 12
10:21:26.478847 IP vm2 > reliance.reliance: ICMP echo reply, id 0, seq 0, length 12
10:21:47.205056 IP vm2 > reliance.reliance: ICMP vm2 udp port echo unreachable, length 37
10:21:47.205600 IP vm2 > reliance.reliance: ICMP vm2 udp port echo unreachable, length 37
10:21:52.172036 IP 20.1.1.1 > 40.1.1.3: ICMP echo request, id 3256, seq 1, length 64
10:21:52.174641 IP 40.1.1.3 > 20.1.1.1: ICMP echo reply, id 3256, seq 1, length 64
10:21:53.173855 IP 20.1.1.1 > 40.1.1.3: ICMP echo request, id 3256, seq 2, length 64
10:21:53.175687 IP 40.1.1.3 > 20.1.1.1: ICMP echo reply, id 3256, seq 2, length 64
^C
11 packets captured
11 packets received by filter
0 packets dropped by kernel
vimansh_mahajan@vm2:~$
```

Gateway interface between gateway and server



```
vimansh_mahajan@vm2:~$ sudo tcpdump -i enp0s8 icmp
[sudo] password for vimansh_mahajan:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s8, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:21:26.478410 IP reliance.reliance > vm2: ICMP echo request, id 0, seq 0, length 12
^C10:21:52.172250 IP 40.1.1.2 > 40.1.1.3: ICMP echo request, id 3256, seq 1, length 64

2 packets captured
5 packets received by filter
0 packets dropped by kernel
vimansh_mahajan@vm2:~$
```

Server (vm4: 40.1.1.3)



```
vimansh_mahajan@vm4:~$ sudo tcpdump -i enp0s3 icmp
[sudo] password for vimansh_mahajan:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
16:37:04.535925 IP vm4 > reliance.reliance: ICMP vm4 udp port echo unreachable, length 37
16:37:09.444060 IP _gateway > vm4: ICMP echo request, id 3256, seq 1, length 64
16:37:09.444193 IP vm4 > _gateway: ICMP echo reply, id 3256, seq 1, length 64
16:37:10.445495 IP _gateway > vm4: ICMP echo request, id 3256, seq 2, length 64
16:37:10.445539 IP vm4 > _gateway: ICMP echo reply, id 3256, seq 2, length 64
^C
5 packets captured
5 packets received by filter
0 packets dropped by kernel
vimansh_mahajan@vm4:~$
```

Request Summary:

- **VM1 (Client: 20.1.1.1)**:
  - **Sends**: ICMP echo request
  - **Before**: Source IP: 20.1.1.1, Destination IP: 40.1.1.3
  - **After**: Source IP: 40.1.1.2, Destination IP: 40.1.1.3 (modified by gateway)
- **VM2 (Gateway: 20.1.1.2)**:
  - **Receives**: ICMP echo request from VM1
  - **Changes Source**: From 20.1.1.1 to 40.1.1.2
  - **Forwards to VM3**: Source IP: 40.1.1.2, Destination IP: 40.1.1.3
- **VM4 (Server: 40.1.1.3)**:
  - **Receives**: ICMP echo request from VM2

Response Summary:

- **VM4 (Server: 40.1.1.3)**:
  - **Sends**: ICMP echo reply
  - **Before Reply**: Source IP: 40.1.1.3, Destination IP: 40.1.1.2
- **VM2 (Gateway: 20.1.1.2)**:
  - **Receives**: ICMP echo reply from VM3
  - **Reverts Destination**: From 40.1.1.2 back to 20.1.1.1
  - **Forwards to VM1**: Source IP: 40.1.1.3, Destination IP: 20.1.1.1
- **VM1 (Client: 20.1.1.1)**:
  - **Receives**: ICMP echo reply from VM2
  - **Final**: Source IP: 40.1.1.3, Destination IP: 20.1.1.1

Q.5 summary

Load balancing at the gateway VM

```
vimansh_mahajan@vm2:~$ sudo iptables -t nat -L PREROUTING -v -n
[sudo] password for vimansh_mahajan:
Chain PREROUTING (policy ACCEPT 375K packets, 34M bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DNAT       0    --  *      *       20.1.1.0/24          20.1.1.2             statistic mode random probability 0.79999999981 to:40.1.1.1
    0     0 DNAT       0    --  *      *       20.1.1.0/24          20.1.1.2             to:40.1.1.3
vimansh_mahajan@vm2:~$
```

Commands:

Route 80% of packets to 40.1.1.1 (higher probability to lower RTT server as per question 3(b)):
sudo iptables -t nat -A PREROUTING -s 20.1.1.1/24 -d 20.1.1.2 -m statistic --mode random --probability 0.8 -j DNAT --to-destination 40.1.1.1

- -t nat: Specifies the NAT table, where Network Address Translation rules are configured.
- -A PREROUTING: Adds this rule to the PREROUTING chain, which alters packets as they arrive before routing.
- -s 20.1.1.1/24: Sets the source IP range to 20.1.1.1/24, meaning traffic originating from 20.1.1.x.
- -d 20.1.1.2: Specifies the destination IP as 20.1.1.2, which is where packets are initially directed.
- -m statistic --mode random --probability 0.8: Uses the statistic module to select 80% of the packets randomly.
- -j DNAT --to-destination 40.1.1.1: Jumps to DNAT (Destination NAT) and changes the destination IP to 40.1.1.1.

Purpose: Redirects 80% of packets from 20.1.1.1 (client) heading to 20.1.1.2 (gateway) to the server at 40.1.1.1.

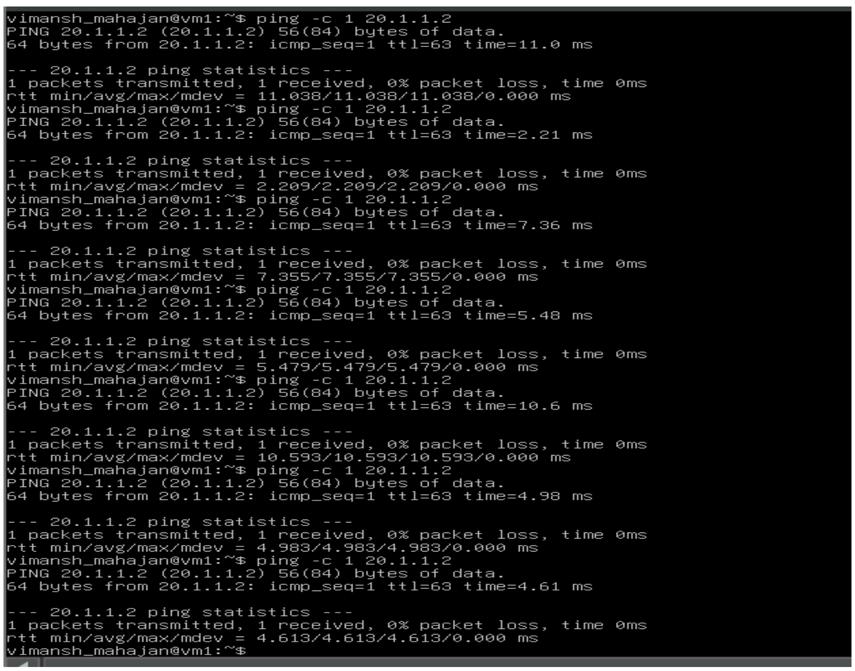
Route remaining packets to 40.1.1.3:

sudo iptables -t nat -A PREROUTING -s 20.1.1.1/24 -d 20.1.1.2 -j DNAT --to-destination 40.1.1.3

- -t nat -A PREROUTING: Adds the rule to the NAT table's PREROUTING chain, affecting packets upon arrival.
- -s 20.1.1.1/24 -d 20.1.1.2: Specifies packets from source 20.1.1.1/24 heading to destination 20.1.1.2.
- -j DNAT --to-destination 40.1.1.3: Jumps to DNAT, setting the destination IP to 40.1.1.3.

Purpose: Redirects any remaining packets from 20.1.1.1 heading to 20.1.1.2 to the server at 40.1.1.3.

Client sends multiple ping requests

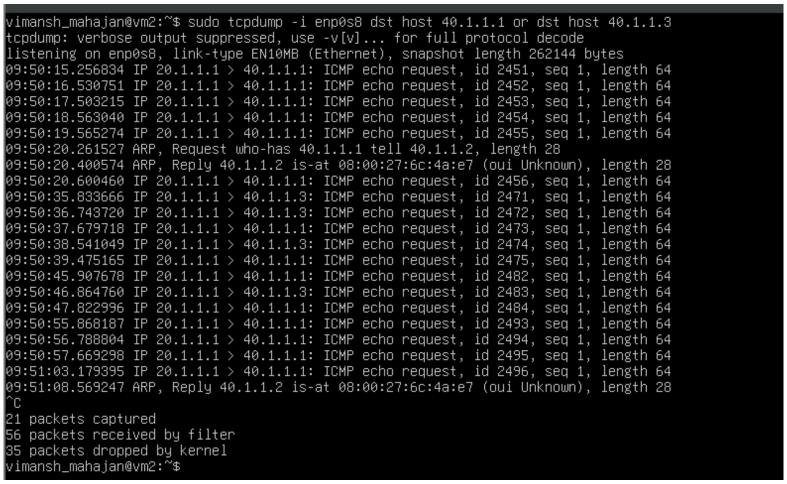


```
vimansh_mahajan@vm1:~$ ping -c 1 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=63 time=11.0 ms

--- 20.1.1.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 11.038/11.038/11.038/0.000 ms
vimansh_mahajan@vm1:~$ ping -c 1 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=63 time=2.21 ms

--- 20.1.1.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2.209/2.209/2.209/0.000 ms
vimansh_mahajan@vm1:~$ ping -c 1 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=63 time=7.36 ms

--- 20.1.1.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 7.355/7.355/7.355/0.000 ms
vimansh_mahajan@vm1:~$ ping -c 1 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=63 time=5.48 ms

--- 20.1.1.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 5.479/5.479/5.479/0.000 ms
vimansh_mahajan@vm1:~$ ping -c 1 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=63 time=10.6 ms

--- 20.1.1.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 10.593/10.593/10.593/0.000 ms
vimansh_mahajan@vm1:~$ ping -c 1 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=63 time=4.98 ms

--- 20.1.1.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 4.983/4.983/4.983/0.000 ms
vimansh_mahajan@vm1:~$ ping -c 1 20.1.1.2
PING 20.1.1.2 (20.1.1.2) 56(84) bytes of data.
64 bytes from 20.1.1.2: icmp_seq=1 ttl=63 time=4.61 ms

--- 20.1.1.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 4.613/4.613/4.613/0.000 ms
vimansh_mahajan@vm1:~$
```

Use tcpdump to monitor traffic

sudo tcpdump -i <interface> <add-on rules>

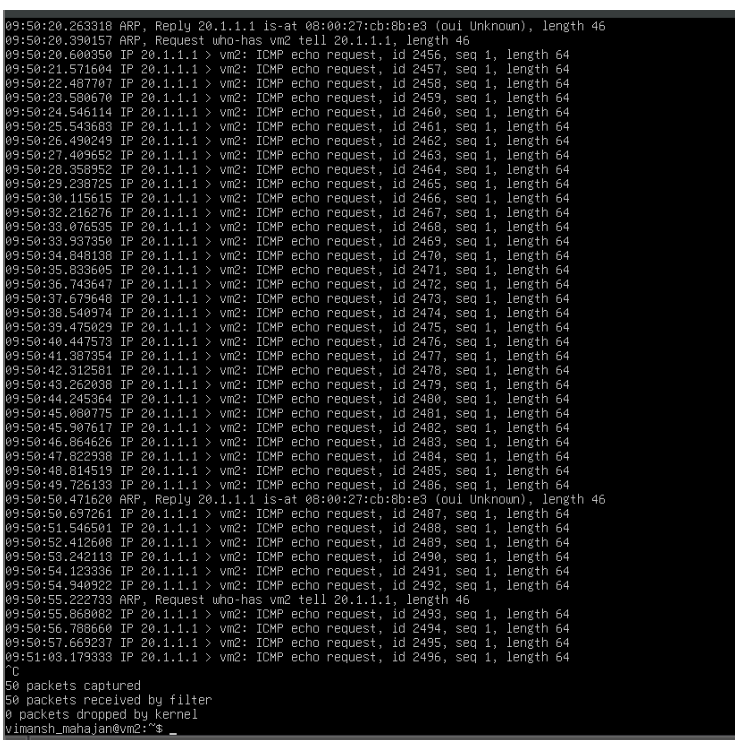
Gateway (interface between gateway and servers)



```
vimansh_mahajan@vm2:~$ sudo tcpdump -i enp0s8 dst host 40.1.1.1 or dst host 40.1.1.3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s8, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:50:15.256834 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2451, seq 1, length 64
09:50:16.530751 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2452, seq 1, length 64
09:50:17.503215 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2453, seq 1, length 64
09:50:18.563040 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2454, seq 1, length 64
09:50:19.565274 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2455, seq 1, length 64
09:50:20.261527 ARP, Request who-has 40.1.1.1 tell 40.1.1.2, length 28
09:50:20.400574 ARP, Reply 40.1.1.2 is-at 08:00:27:6c:4a:e7 (oui Unknown), length 28
09:50:20.600460 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2456, seq 1, length 64
09:50:35.833666 IP 20.1.1.1 > 40.1.1.3: ICMP echo request, id 2471, seq 1, length 64
09:50:36.743720 IP 20.1.1.1 > 40.1.1.3: ICMP echo request, id 2472, seq 1, length 64
09:50:37.679718 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2473, seq 1, length 64
09:50:38.541049 IP 20.1.1.1 > 40.1.1.3: ICMP echo request, id 2474, seq 1, length 64
09:50:39.475165 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2475, seq 1, length 64
09:50:45.907678 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2482, seq 1, length 64
09:50:46.864760 IP 20.1.1.1 > 40.1.1.3: ICMP echo request, id 2483, seq 1, length 64
09:50:47.822996 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2484, seq 1, length 64
09:50:55.868187 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2493, seq 1, length 64
09:50:56.788804 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2494, seq 1, length 64
09:50:57.669298 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2495, seq 1, length 64
09:51:03.179395 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2496, seq 1, length 64
09:51:08.569247 ARP, Reply 40.1.1.2 is-at 08:00:27:6c:4a:e7 (oui Unknown), length 28
^C
21 packets captured
56 packets received by filter
35 packets dropped by kernel
vimansh_mahajan@vm2:~$
```

As visible in the screenshot, a few requests are sent to 40.1.1.3 while many others are routed to 40.1.1.1 which is in line with the 80-20 probability rules set.

Gateway (interface between client and gateway)



```
09:50:20.263318 ARP, Reply 20.1.1.1 is-at 08:00:27:cb:8b:e3 (oui Unknown), length 46
09:50:20.390157 ARP, Request who-has vm2 tell 20.1.1.1, length 46
09:50:20.600350 IP 20.1.1.1 > vm2: ICMP echo request, id 2456, seq 1, length 64
09:50:21.571604 IP 20.1.1.1 > vm2: ICMP echo request, id 2457, seq 1, length 64
09:50:22.487707 IP 20.1.1.1 > vm2: ICMP echo request, id 2458, seq 1, length 64
09:50:23.580670 IP 20.1.1.1 > vm2: ICMP echo request, id 2459, seq 1, length 64
09:50:24.546114 IP 20.1.1.1 > vm2: ICMP echo request, id 2460, seq 1, length 64
09:50:25.543683 IP 20.1.1.1 > vm2: ICMP echo request, id 2461, seq 1, length 64
09:50:26.490249 IP 20.1.1.1 > vm2: ICMP echo request, id 2462, seq 1, length 64
09:50:27.409652 IP 20.1.1.1 > vm2: ICMP echo request, id 2463, seq 1, length 64
09:50:28.358952 IP 20.1.1.1 > vm2: ICMP echo request, id 2464, seq 1, length 64
09:50:29.238725 IP 20.1.1.1 > vm2: ICMP echo request, id 2465, seq 1, length 64
09:50:30.115615 IP 20.1.1.1 > vm2: ICMP echo request, id 2466, seq 1, length 64
09:50:32.216276 IP 20.1.1.1 > vm2: ICMP echo request, id 2467, seq 1, length 64
09:50:33.076535 IP 20.1.1.1 > vm2: ICMP echo request, id 2468, seq 1, length 64
09:50:33.937350 IP 20.1.1.1 > vm2: ICMP echo request, id 2469, seq 1, length 64
09:50:34.848138 IP 20.1.1.1 > vm2: ICMP echo request, id 2470, seq 1, length 64
09:50:35.833605 IP 20.1.1.1 > vm2: ICMP echo request, id 2471, seq 1, length 64
09:50:36.743647 IP 20.1.1.1 > vm2: ICMP echo request, id 2472, seq 1, length 64
09:50:37.679648 IP 20.1.1.1 > vm2: ICMP echo request, id 2473, seq 1, length 64
09:50:38.540974 IP 20.1.1.1 > vm2: ICMP echo request, id 2474, seq 1, length 64
09:50:39.475029 IP 20.1.1.1 > vm2: ICMP echo request, id 2475, seq 1, length 64
09:50:40.447573 IP 20.1.1.1 > vm2: ICMP echo request, id 2476, seq 1, length 64
09:50:41.387354 IP 20.1.1.1 > vm2: ICMP echo request, id 2477, seq 1, length 64
09:50:42.312581 IP 20.1.1.1 > vm2: ICMP echo request, id 2478, seq 1, length 64
09:50:43.262038 IP 20.1.1.1 > vm2: ICMP echo request, id 2479, seq 1, length 64
09:50:44.245364 IP 20.1.1.1 > vm2: ICMP echo request, id 2480, seq 1, length 64
09:50:45.080775 IP 20.1.1.1 > vm2: ICMP echo request, id 2481, seq 1, length 64
09:50:45.907617 IP 20.1.1.1 > vm2: ICMP echo request, id 2482, seq 1, length 64
09:50:46.864626 IP 20.1.1.1 > vm2: ICMP echo request, id 2483, seq 1, length 64
09:50:47.822938 IP 20.1.1.1 > vm2: ICMP echo request, id 2484, seq 1, length 64
09:50:48.814519 IP 20.1.1.1 > vm2: ICMP echo request, id 2485, seq 1, length 64
09:50:49.726133 IP 20.1.1.1 > vm2: ICMP echo request, id 2486, seq 1, length 64
09:50:50.471620 ARP, Reply 20.1.1.1 is-at 08:00:27:cb:8b:e3 (oui Unknown), length 46
09:50:50.697261 IP 20.1.1.1 > vm2: ICMP echo request, id 2487, seq 1, length 64
09:50:51.546501 IP 20.1.1.1 > vm2: ICMP echo request, id 2488, seq 1, length 64
09:50:52.412608 IP 20.1.1.1 > vm2: ICMP echo request, id 2489, seq 1, length 64
09:50:53.242113 IP 20.1.1.1 > vm2: ICMP echo request, id 2490, seq 1, length 64
09:50:54.123336 IP 20.1.1.1 > vm2: ICMP echo request, id 2491, seq 1, length 64
09:50:54.940922 IP 20.1.1.1 > vm2: ICMP echo request, id 2492, seq 1, length 64
09:50:55.222733 ARP, Request who-has vm2 tell 20.1.1.1, length 46
09:50:55.868082 IP 20.1.1.1 > vm2: ICMP echo request, id 2493, seq 1, length 64
09:50:56.788660 IP 20.1.1.1 > vm2: ICMP echo request, id 2494, seq 1, length 64
09:50:57.669237 IP 20.1.1.1 > vm2: ICMP echo request, id 2495, seq 1, length 64
09:51:03.179333 IP 20.1.1.1 > vm2: ICMP echo request, id 2496, seq 1, length 64
^C
50 packets captured
50 packets received by filter
0 packets dropped by kernel
vimansh_mahajan@vm2:~$ _
```

Server 40.1.1.1

```
vimansh_mahajan@vm3:~$ sudo tcpdump -i enp0s3 src host 20.1.1.1
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
18:06:31.160843 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2451, seq 1, length 64
18:06:32.433890 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2452, seq 1, length 64
18:06:33.406200 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2453, seq 1, length 64
18:06:34.467493 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2454, seq 1, length 64
18:06:35.468289 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2455, seq 1, length 64
18:06:36.503811 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2456, seq 1, length 64
18:06:38.386270 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2458, seq 1, length 64
18:06:39.477461 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2459, seq 1, length 64
18:06:48.101520 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2467, seq 1, length 64
18:06:48.960608 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2468, seq 1, length 64
18:06:49.820756 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2469, seq 1, length 64
18:06:50.730322 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2470, seq 1, length 64
18:06:53.559352 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2473, seq 1, length 64
18:06:55.353503 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2475, seq 1, length 64
18:06:56.325247 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2476, seq 1, length 64
18:06:57.264205 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2477, seq 1, length 64
18:06:58.190548 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2478, seq 1, length 64
18:06:59.137388 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2479, seq 1, length 64
18:07:00.955480 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2481, seq 1, length 64
18:07:01.781235 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2482, seq 1, length 64
18:07:03.695393 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2484, seq 1, length 64
18:07:04.686364 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2485, seq 1, length 64
18:07:05.597432 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2486, seq 1, length 64
18:07:07.416621 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2488, seq 1, length 64
18:07:08.282435 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2489, seq 1, length 64
18:07:09.111685 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2490, seq 1, length 64
18:07:09.991851 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2491, seq 1, length 64
18:07:11.735793 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2493, seq 1, length 64
18:07:12.655868 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2494, seq 1, length 64
18:07:13.535891 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2495, seq 1, length 64
18:07:19.044939 IP 20.1.1.1 > 40.1.1.1: ICMP echo request, id 2496, seq 1, length 64
```

Server 40.1.1.3

```
vimansh_mahajan@vm4:~$ sudo tcpdump -i enp0s3 src host 20.1.1.1
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
18:06:33.286195 IP 20.1.1.1 > vm4: ICMP echo request, id 2457, seq 1, length 64
18:06:40.063351 IP 20.1.1.1 > vm4: ICMP echo request, id 2464, seq 1, length 64
18:06:47.530364 IP 20.1.1.1 > vm4: ICMP echo request, id 2471, seq 1, length 64
18:06:48.439011 IP 20.1.1.1 > vm4: ICMP echo request, id 2472, seq 1, length 64
18:06:50.234699 IP 20.1.1.1 > vm4: ICMP echo request, id 2474, seq 1, length 64
18:06:55.934886 IP 20.1.1.1 > vm4: ICMP echo request, id 2480, seq 1, length 64
18:06:58.552192 IP 20.1.1.1 > vm4: ICMP echo request, id 2483, seq 1, length 64
18:07:02.382749 IP 20.1.1.1 > vm4: ICMP echo request, id 2487, seq 1, length 64
18:07:06.623876 IP 20.1.1.1 > vm4: ICMP echo request, id 2492, seq 1, length 64
```

To 40.1.1.3: 9 packets were sent (counted from the given screenshot)

To 40.1.1.1: 31 packets were sent (counted from the given screenshot)

Therefore total packets = 40

40.1.1.3 got: (9/40)*100 = 22.5 % **(almost 20 % packets)**

40.1.1.1 got: (31/40)*100 = 77.5% **(almost 80% packets)**

This shows that 80 : 20 load balancing works